

RITESH SETH

Curriculum Vitae

B-513, 5th Floor, R&D Building
IIIT Delhi, New Delhi, India
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Research Experience

Aug.'19–
Dec.'21 **Research Project in Algorithm and Computational Geometry**, *Indian Statistical Institute, Kolkata (India)*, supervisor : Prof. Subhas C. Nandy, (Focused on : Minimum height watchtower on terrain, Shortest path with obstacle violation, 1-violation path in simple polygon)

Education

2022–Cont. **PhD. in Computer Science**, *Indraprastha Institute of Information Technology, Delhi (India)*
2017–2019 **M.Sc. in Computer Science**, *Ramakrishna Mission Vivekananda Educational and Research Institute, Belur Math, Howrah, West Bengal (India)*
2014–2017 **B.Sc.(hons.) in Computer Science**, *University of Calcutta, Kolkata (India)*
2012–2014 **Higher Secondary Examination (10+2)**, *West Bengal Council of Higher Secondary Education, India*
2010–2012 **Secondary Examination**, *West Bengal Board of Secondary Education, India*

Theses

masters **A survey on homomorphisms of signed graphs**, supervisors: Prof. Sagnik Sen and Prof. Soumen Nandi

Publications

conference **"Efficient algorithms for Incremental Metric Bipartite Matching"**, Ritesh Seth, Mrinal Garg, Sujoy Bhore, Sharath Raghvendra and Syamantak Das, ICLR, 2026
journal **"Acrophobic guard watchtower problem"**, Ritesh Seth, Anil Maheshwari and Subhas C Nandy, Computational Geometry, 2023
conference **"An Acrophobic Guard Watchtower Problem on Terrains."**, Ritesh Seth, Anil Maheshwari and Subhas C Nandy, CCCG, 2021
conference **"The Relative Signed Clique Number of Planar Graphs is 8."**, Sandip Das, Soumen Nandi, Sagnik Sen and Ritesh Seth., CALDAM, 2019

Teaching Experience

Aug.'25–
Dec.'25 **Working as a TA in Usable Security and Privacy (CSE547)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)
Jan.'25–
May.'25 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)

- Aug.'24– Dec.'24 **Worked as the TA in Modern Algorithm Design (CSE519)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)
- Jan.'24– May.'24 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)
- Aug.'23– Dec.'23 **Worked as the TA in Modern Algorithm Design (CSE519)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)
- Jan.'23– May.'23 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)
- Aug.'22– Dec.'22 **Worked as a TA in Introduction to Graduate Algorithms (CSE525)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)
- Jan.'22– May.'22 **Worked as a TA in Analysis and Design of Algorithms (CSE222)**, Department of Computer Science and Engineering
Indraprastha Institute of Information Technology Delhi (IIITD)
- Jan.'20– May.'20 **Information Technology in Sports (MSSCT-206)**, (*Basics of Python, basics of ML using Python, LaTeX*), M.Sc sports 19-21 batch, Department of Sports Science and Yoga
Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)
- Jul.'19– Nov.'19 **Computer Application (PGDY-CP-108)**, (*Basics of Computer Applications including Word, Excel and PowerPoint*), PGDY 2019-20 batch, Department of Sports Science and Yoga
Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI)

Projects and Internships

- Aug.'19– Dec.'21 **Research Project**, *Indian Statistical Institute, Kolkata (India)*, supervisor : Prof. Subhas C. Nandy
- Feb.–Mar.'19 **Research Internship**, *Indian Institute of Technology, Dharwad (India)*, supervisor : Prof. Sagnik Sen
- Dec.'18– Jan.'19 **Research Internship**, *Indian Statistical Institute, Kolkata (India)*, supervisor : Prof. Soumen Nandi
- Sep.'18 **Presentation in open problem session in "ELGGA"**, *BITS Pilani, Hyderabad (India) in association with ACSDM (Association for Computer Science and Discrete Mathematics)*
- June.'18 **Summer Project**, *Ramakrishna Mission Vivekananda Educational and Research Institute, Howrah (India)*, supervisor : Prof. Sagnik Sen

Funding / Scholarships

- June.'19 **Obtained funding for being TA at ACM summer school on "Geometric Algorithms and Their Applications"**, *NISER, Bhubaneswar (India)*
- Feb.–Mar.'19 **Obtained funding for Research Internship**, *Indian Institute of Technology, Dharwad (India)*, supervisor : Prof. Sagnik Sen
- Sep.'18 **Was selected and obtained funding for "ELGGA"**, *BITS Pilani, Hyderabad (India) in association with ACSDM (Association for Computer Science and Discrete Mathematics)*
- 2017-19 **Swami Vivekananda Merit-Cum-Means Scholarship**, *Government of West Bengal, India*

Technical Skills

Programming Languages

C, C++, Python, CUDA, Visual Basic, Visual C++

Tools

Scikit-Learn, Keras, PyTorch

Workshop / Conference Attended

- April.'26 **The Fourteenth International Conference on Learning Representations (ICLR)**, *Rio-centro Convention and Event Center, Rio de Janeiro (Brazil)*, Presenter (Efficient algorithms for Incremental Metric Bipartite Matching)
- Dec.'24 **Frontiers of Geometric Algorithms**, *IISc, Bengaluru (India)*
- Oct.'24 **An ACM IndiCS Seminar on “Continuous Methods in Discrete Optimization and Complexity”**, *Infosys Mysore campus, Karnataka (India)*
- July.'23 **Recent Trends in Algorithms**, *NISER, Bhubaneswar (India)*
- Dec.'22 **Foundations of Software Technology and Theoretical Computer Science (FSTTCS) on Algorithms under Uncertainty**, *IIT, Madras (India)*
- Aug.'21 **33rd Canadian Conference on Computational Geometry (CCCG)**, *Halifax, Nova Scotia (Canada)*, Speaker (An Acrophobic Guard Watchtower Problem on Terrains)
- June.'19 **ACM summer school on “Geometric Algorithms and Their Applications”**, *NISER, Bhubaneswar (India)*, Organizing Committee-Volunteer
- Jan.'19 **Series of lectures: AGraHo Project**, *Organized by Ramakrishna Mission Vivekananda Educational and Research Institute, Belur Math, Howrah (India)*
- Feb.'18 **Workshop on Big Data Analytics**, *Ramakrishna Mission Vivekananda Educational and Research Institute, Belur Math, Howrah (India)*, Organizing Committee-Volunteer